

MAT 380 Final Project

Your goal for this project is to explore an analysis topic beyond what we have done in class and teach it to your fellow classmates. The #1 goal is to understand it well enough that you can teach it to others.

You will present your topic in a **25-minute presentation** in the final exam period of class. Your primary goal should be understandability. You are free to gloss over details that are not necessary for understanding, but are encouraged to include proofs that are especially illuminating (also consider presenting a proof outline rather than a full proof). You are also encouraged to use examples to illustrate the concepts.

- You should put together a **Beamer (LaTeX) presentation**. I am requiring LaTeX because the math formatting is abysmal in PowerPoint and nonexistent in Google Slides. I suggest finding a template on Overleaf that you like.
- You may work alone or in pairs of your choosing on your final project and presentation. **By Wednesday, April 15, fill out the Final Project Selection Form** on Moodle (rank your topic choices and solidify your groups).
- Normally on homework, I do not want you consulting the internet for help/looking up proofs. On this project, you are explicitly allowed to do so as long as you internalize the proofs and complete them in your own way (not plagiarizing). Please list links of any sources that you use.

Topics

Not all of these topics are “equal in difficulty.” Some are direct extensions of definitions we’ve worked with in this class, and some are more out-there explorations. If you are doing a more direct extension, there is more of a standard for the rigor of your arguments. See below for details.

**If, in the process of putting together your presentation, you find that your topic as stated is too ambitious and includes too much for a 25-minute presentation, come talk to me and propose how you’d like to trim it down.*

Direct Extensions

For direct extensions, you should take care in your presentation to present and thoroughly explain the rigorous definitions and theorem statements involved in your topic. You should also include at least one full proof in your presentation that feels important, illuminating, and standard to the way things are proved regarding your topic.

1. Convergence of Infinite Series (Section 2.4, 2.7)

Choose this if you liked infinite series from calculus, and convergence in this course.

Present the definition of convergence of an infinite series. Explain how our proof of divergence of the harmonic series generalizes to the Cauchy Condensation Test. Present the Cauchy Criterion for Series and explain how it easily proves Theorem 2.7.3 and the Comparison Test. Finally, present facts about geometric series to prove the p-series (Corollary 2.4.7, Exercise 2.7.5).

2. Baire Category Theorem (Section 3.5, 8.2)

Choose this if you liked topology.

Read section 3.5 and present the new topological definitions of F_σ sets, G_δ sets, and nowhere-dense sets. Prove Baire’s Theorem (3.5.4). Explain how these definitions and results can be generalized to

a general metric space (section 8.2), and discuss the implications of the Baire Category Theorem (in the following paragraph).

3. Uniform Continuity and Continuous Extension (Section 4.4)

Choose this if you liked continuity.

Read the section about Uniform Continuity and present this new definition and how it differs from regular continuity. Discuss uniformly continuous functions on compact sets (Theorem 4.4.7). Furthermore, discuss the Continuous Extension Theorem as it relates to Cauchy Sequences in Exercise 4.4.13.

4. Sets of Discontinuity (Section 4.6)

Choose if you liked topology and continuity.

Follow the outline in the book and prove that the set of discontinuity D_f for any function must be a countable union of closed sets. Furthermore, read and summarize the importance of this statement in Epilogue Section 4.7. (If someone chooses the Baire Category Theorem, you may want to talk to them.)

5. Sequences of Functions (Section 6.2)

Choose this if you liked convergence.

Discuss the definitions of pointwise convergence and uniform convergence of a sequence of functions. Give an example of a sequence of functions that converges pointwise but not uniformly to illustrate the difference. Prove the Cauchy Criterion for Uniform Convergence, and discuss the Continuous Limit Theorem.

Out-There Explorations

If you are doing an exploration, you are free to focus less on rigor in the pursuit of understanding and presentation (you can get away with assuming some things without proof). Sometimes you will run into things we haven't talked about. Your job is to understand the idea without getting lost in the details. You should not feel the need to write every detail, rather only write the details necessary for the idea, and make sure that you understand any details you choose to include in your presentation.

6. A Continuous, Nowhere-Differentiable Function and Brief Look at Fourier Series (Section 5.4, 8.5)

Choose this if you enjoyed nonintuitive functions with weird properties.

Read Section 5.4 carefully and present this class of nowhere-differentiable functions. (You may want to talk to the people doing convergence of infinite series.) Briefly explain why this is a "specific example of Fourier series" and discuss what the point/main ideas behind Fourier series are without getting too lost in the weeds. Work through the exercises in this chapter to prove that this class of functions is continuous but nowhere-differentiable.

7. Weierstrass Approximation Theorem and a Refresher on Taylor Series (Section 6.7)

Choose this if you like "crunchy" constructive proofs of existence, and the idea of approximation.

Read Section 6.7 carefully and work through the proof of WAT and the ideas of interpolating polygonal functions. You will likely need to find other sources, as a lot of the proof is left as rather difficult exercises. Use this as an opportunity to refresh our memory on how Taylor Series work (without rigorous proof) and how they are used in this proof. Make sure and answer Exercise 6.7.9.

8. Functional Analysis and the Metric Space of Continuous Functions $C[0, 1]$ (Section 8.2)

Choose this if you would like to explore more with how everything we've talked about generalizes to a different metric space.

Discuss the metric space of continuous functions on $[0, 1]$, $C[0, 1]$. Explore various metrics in Exercise 8.2.2 and explain why it is complete under the "sup norm" (8.2.5b) - you should read a proof of this instead of trying to prove it yourself because it involves some results we haven't talked about. Consider functions from this space in Exercise 8.2.6, as well as open and closed sets of the space in 8.2.9. Give a very general outline of the proof of Theorem 8.2.12 and explain its implications (If someone chooses the Baire Category Theorem, you may want to talk to them.)

9. Existence of the Real Numbers (Section 8.6)

Choose this if you enjoy the absurdity of proving what seems to be obvious. It may also help if you took Abstract Algebra (deals with definition of an ordered field).

First, discuss why we would even need to try and prove the existence of real numbers. Read section 8.6, and discuss both Dedekind's approach and Cantor's approach for constructing $\mathbb{R}$ from $\mathbb{Q}$. Discuss how Dedekind's approach leads to $\mathbb{R}$ being considered an ordered field (discuss what this means, not necessarily proving everything). Using this, prove why Dedekind's approach leads to $\mathbb{R}$ being complete.

***Even though this is definitely considered an "out-there extension," you should still treat this topic with quite a bit of rigor, as the mere thought of proving real numbers exist seems like something we should do rigorously if we do it at all. Provide at least one full proof in your presentation.*

10. Lebesgue Measure and Integration

***This project will use the book "A User-Friendly Introduction to Lebesgue Measure and Integration" by Gail S. Nelson. See me to borrow my copy.*

Choose this if you liked learning about integration, and are wondering if there are other ways to do it. Involves thinking similar to our idea of "open covers."

Discuss the definition of Lebesgue outer measure (Def. 1.1.1), and include examples of sets and their Lebesgue measure (including the Cantor Set on p. 21). Discuss the definition of Lebesgue integral (Def 2.25), and discuss how it differs from the Riemann integral. Explain why Dirichlet's function is not Riemann Integrable, but is Lebesgue Integrable (Examples 2.2.6, 2.2.9).

Notes: The book defines measure on a subset of $\mathbb{R}^n$, please restrict your discussion to $\mathbb{R}$. In general, the book goes into a lot of detail, your job is to provide an understandable quick overview.

Topics Not on this List

Topics not on this list are allowed, but will need a proposal on the Final Project Selection Form due April

15. Topic proposals should include:

- Whether you think it's a "direct extension" or "out-there exploration"
- What definitions/theorems/examples you are going to present
- What results/definitions from this course you are going to be using
- At least one source that you will be consulting (make sure I have access to it. If it is a book, scan the pages for me that you will be consulting)